

Roll No.

41478

**B. Pharm. 4th Semester
(Regular/Re-appear/Impr.)
Examination – May, 2024
PHYSICAL PHARMACEUTICS-II**

Paper : BP403T

Time : Three hours]

[Maximum Marks : 75

Before answering the questions, candidates should ensure that they have been supplied the correct and complete question paper. No complaint in this regard, will be entertained after examination.

Note : Attempt *five* questions in all, selecting *one* question from each Unit. Question No. 1 is *compulsory* consisting of 8 short answer questions. All question carry equal marks.

1. Short answers on: (Answer all the questions) :

$10 \times 2 = 20$

1. Write the relationship among true solution, coarse suspension and colloids.
2. Define light powders and heavy powders with example.
3. Write Gold number with its Significance.

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4. Zero order reactions are dependent upon the concentration of reactant. Comment.
5. What is plug flow ? How can it be minimized ?
6. What are Bulges and Spurs ?
7. What is glidant ? Write two suitable examples of it.
8. Define Rheology. Give any two applications.
9. State electro-kinetic and electro-dynamic in colloids
10. Describe carr's index

Long Answers (Answer 2 out of 3)

$2 \times 10 = 20$

2. Write the principle, construction and working of Coulter-Counter apparatus for the determination of particle size of powders. 10
3. Define Non-Newtonian flow of liquid. Describe the principle, construction of cup-bob over the advantage point in cone-plate method. 10
4. Define how shelf life of a medicinal product is determined. 10

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Short Answers (Answer 7 out of 9)

$7 \times 5 = 35$

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5. Explain kinetic properties of colloids
 6. Discuss plastic and pseudoplastic system of flow.
 7. Write HLB scale showing classification of surfactant function
 8. Explain Donnan membrane effect with suitable example
 9. Explain briefly different types of tests to identify the types of emulsion.
 10. State Edmundson's equation.
 11. Give expressions for rate constant and half-life of first order reaction.
 12. What is specific surface area ? (How is it measured by air permeability method ?)
 13. Differentiate between flocculated and deflocculated suspensions.
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Department of Pharmaceutical Sciences, Indira Gandhi University, Meerpur, Rewari
B. Pharmacy 4th Semester 2nd Sessional Examination

Physical Pharmaceutics(Theory)(Paper code –BP 403T)

Duration- 1h 30 min

Max. Marks- 30

Q1 is compulsory. Each part carries 2 marks.

Define the following:

- i. Stress
- ii. Strain
- iii. Elastic modulus
- iv. Thixotropy
- v. Dispersed system

Attempt any two out of three. Each question carries 5 marks.

Q2. Explain Newtonian System.

Q3. . Explain Non-Newtonian System.

Q4. Plastic and Elastic Deformation.

Attempt any one out of two. Each question carries 10 marks.

Q5. Define size & shapes of colloidal particles. Classification of colloids.

Q6. Determination of viscosity by using Ostwald's viscometer.

*Molecular
colloids
Association*

Indira Gandhi University, Meerpur, Rewari
Department of Pharmaceutical Sciences
B.Pharma 2nd Semester, 1st Sessional Examination, 2024
Paper Code -BP403T physical Pharmaceutics (Theory)

Duration: 1 hr 30 minutes

Question no.1 is compulsory

Total marks: 30

Parameter

- a. Micro emulsion
- b. Suspension
- c. Particle shape
- d. Specific surface
- e. Adsorption

2x5=10 marks

2 Attempt any one questions from the rest

- a. Define method to determine surface area
- b. Define methods to determine particle size ✓

10x1=10 marks

3 Attempt any two questions from the rest

- a. Explain density. Types of density
- b. Define derived properties of powder (5) -
- c. Emulsion formulation by HLB method

5x2=10 marks

- Bulk density
- Tapped density
- Particle size

GD
BO
TO

- Pass
Optical
Density
Surface
Area
Particle
Size
Concentration
Hygroscopic